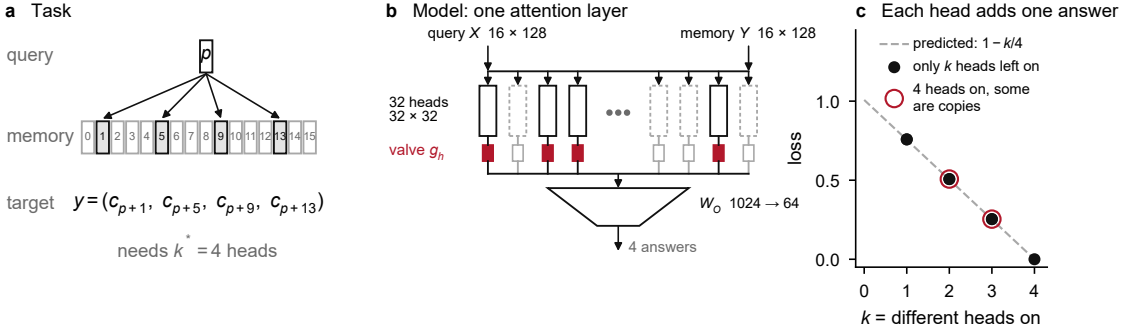
a Task
query
p
memory
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
target $y = (c_{p+1}, c_{p+5}, c_{p+9}, c_{p+13})$
needs $k^* = 4$ heads
b Model: one attention layer
query X 16 × 128
memory Y 16 × 128
32 heads
32 × 32
valve g_h
W_O 1024 → 64
4 answers
c Each head adds one answer
predicted: 1 − k/4
only k heads left on
4 heads on, some are copies
loss
1.0
0.5
0.0
0 1 2 3 4
k = different heads on